

Adaptive Delay-Window and Soft Low-Rank Denoising for MIMO-OFDM Channel Estimation

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ABSTRACT Accurate channel estimation is essential for achieving high spectral efficiency in multi-antenna wireless communication systems. However, least-squares channel estimates are typically corrupted by noise, which degrades the performance of precoding algorithms. In this work, we investigate denoising of MIMO-OFDM channel estimates in order to improve the performance of singular-value decomposition (SVD)-based precoding. A baseline delay-domain hard-window filter is first implemented, which exploits the limited delay spread of wireless propagation. Motivated by structural properties of MIMO channels and recent transform-domain denoising approaches, an adaptive denoising algorithm is proposed that combines noise-floor-corrected delay support estimation with soft low-rank shrinkage of the retained delay-domain channel matrices. The performance of the proposed method is evaluated using the instantaneous spectral-efficiency metric and empirical cumulative distribution functions (CDFs) across all time-frequency samples. Experiments are performed for uplink channel-estimation SNR values of -20 dB, -15 dB, and -10 dB, while the downlink SNR is fixed at 5 dB. The proposed method demonstrates consistent improvements over the hard-window baseline across all tested SNR regimes. Structural analysis using the power delay profile, beam-domain spectrum, and singular value distribution supports the assumptions underlying the proposed algorithm. The results indicate that exploiting both delay sparsity and approximate low-rank spatial structure can improve channel denoising performance and consequently enhance spectral efficiency.

I. INTRODUCTION

Modern wireless communication systems rely heavily on multi-antenna technologies such as multiple-input multiple-output (MIMO) transmission in order to increase data rates and spectral efficiency. In such systems, accurate channel state information is essential because beamforming and precoding algorithms depend directly on the estimated channel matrix. In practice, channel estimation is typically performed using pilot signals and simple least-squares (LS) estimation techniques. While LS estimation is computationally efficient, it produces noisy channel estimates, especially in low signal-to-noise ratio (SNR) conditions. When these noisy estimates are used to design precoders, the resulting beam directions become inaccurate, which can significantly degrade the achievable spectral efficiency.

One common approach to improve channel estimates is to exploit structural properties of wireless propagation channels. In orthogonal frequency division multiplexing (OFDM) systems, the channel impulse response typically has a limited delay spread. As a result, the delay-domain representation of the channel contains significant energy only within a small number of delay taps. This property enables simple denoising techniques such as delay-domain hard-window filtering, where channel taps outside the expected delay support are set to zero.

However, delay-domain filtering alone does not fully exploit all structural properties of MIMO channels. In many

wireless environments, the spatial channel matrix also exhibits approximate low-rank structure due to the limited number of dominant propagation paths. This observation motivates the development of denoising methods that incorporate both delay sparsity and spatial low-rank properties.

In this report, we study channel denoising for SVD-based MIMO precoding. We first implement a delay-domain hard-window baseline method. Then we propose a denoising approach that combines adaptive delay-support estimation in the delay domain with soft low-rank shrinkage of the retained channel matrices. The proposed algorithm remains computationally simple while exploiting additional channel structure beyond the baseline method. The main contributions of this work are summarized as follows:

1. Implementation of a spectral-efficiency evaluation framework based on instantaneous rate and empirical CDF analysis.
2. Development of an adaptive delay-window plus soft low-rank channel denoising algorithm.
3. Performance evaluation across multiple uplink SNR conditions.
4. Interpretation of the algorithm using channel structural analysis including power delay profile, beam-domain spectrum, and singular value distribution.

II. Channel Denoising Methodology and Literature Context

A. Challenges in Noisy MIMO Channel Estimation

In practical wireless systems, the channel state information available at the transmitter is typically obtained through pilot-based channel estimation. A commonly used estimator is the least-squares (LS) estimator because of its simplicity and low computational complexity. However, LS estimation does not exploit any structural properties of the wireless channel and therefore produces noisy channel estimates, especially when the uplink signal-to-noise ratio (SNR) is low.

The presence of estimation noise directly affects the design of precoding matrices. In SVD-based precoding systems, the transmitter constructs beamforming directions from the singular vectors of the estimated channel matrix. If the channel estimate is corrupted by noise, the singular vectors become inaccurate, leading to suboptimal beamforming directions and reduced spectral efficiency.

Consequently, denoising channel estimates before precoder computation can significantly improve system performance.

B. Structural Properties of Wireless Channel

Wireless propagation channels exhibit several structural characteristics that can be exploited for denoising.

1. Limited Delay Spread:

In OFDM systems, the channel can be represented in the delay domain by applying an inverse Fourier transform across subcarriers. Due to the finite propagation delays of multipath components, the channel impulse response typically contains significant energy only within a small number of delay taps.

Therefore, many delay-domain coefficients contain mostly noise and can be safely removed. This observation motivates delay-domain filtering techniques such as hard-window filtering.

2. Directional Propagation and Beam-Domain Structure:

In large antenna arrays, signals typically arrive from a limited number of spatial directions. When the channel is transformed into the beam domain using spatial Fourier transforms, the channel energy becomes concentrated in a few angular bins corresponding to dominant propagation paths.

Recent research has proposed beam-domain and beam-delay domain channel models that exploit this directional sparsity

to improve channel estimation accuracy. These models jointly exploit correlations across antennas and subcarriers to achieve better noise suppression.

Although such approaches can achieve excellent performance, they often involve complex inference algorithms and high computational cost.

3. Approximate Low-Rank Spatial Structure:

Another important property of MIMO channels is their approximate low-rank behavior. In many wireless environments, the number of dominant propagation paths is limited. As a result, the spatial channel matrix can often be well approximated using only a small number of dominant singular values.

This observation suggests that weaker singular components may contain a large proportion of noise and can be removed without significantly affecting the useful channel information.

Low-rank approximation techniques therefore provide an effective mechanism for suppressing noise in channel matrices.

C. Literature Overview of Channel Denoising Methods

Channel denoising techniques for MIMO systems can generally be grouped into three categories.

1. Delay-Domain Filtering:

The simplest and most widely used method exploits delay sparsity. By transforming the channel across the frequency dimension into the delay domain, taps outside the expected delay spread can be removed. This approach is computationally efficient and forms the baseline method in this work.

However, delay-domain filtering only exploits one dimension of channel structure and does not address spatial noise within the retained taps.

2. Beam-Domain and Beam-Delay Domain Methods:

More advanced techniques transform the channel into the beam domain or joint beam-delay domain. In these representations, channel energy often becomes sparse because multipath components arrive from a limited number of spatial directions.

Several recent works propose algorithms that jointly exploit sparsity across antennas and subcarriers. These methods

typically rely on compressed sensing or message passing techniques to recover sparse channel representations.

While these methods can achieve high accuracy, they are often computationally expensive and require detailed modeling of the propagation environment.

3. Low-Rank Channel Denoising:

Low-rank approximation methods operate directly on the spatial channel matrix. By performing singular value decomposition (SVD), the channel matrix can be decomposed into orthogonal spatial modes with associated singular values.

Because the largest singular values correspond to dominant propagation paths, truncating smaller singular values provides a natural way to suppress noise.

Low-rank denoising has the advantage of being relatively simple to implement while still exploiting important spatial structure.

III. Motivation for the Proposed Method

The proposed algorithm in this work is motivated by combining two complementary structural properties of wireless channels:

1. **Delay sparsity**, which indicates that only a limited number of delay taps contain significant channel energy.

2. **Spatial low-rank structure**, which indicates that most of the channel energy is concentrated in a small number of spatial modes.

The baseline hard-window method exploits only delay sparsity. However, even within the retained delay taps, the spatial channel matrices may still contain significant noise.

To address this limitation, the proposed method first estimates the dominant delay support adaptively from the noisy power delay profile after noise-floor correction. It then applies soft low-rank shrinkage to the spatial channel matrices within the retained delay support.

This approach provides an effective compromise between performance and computational complexity. Compared to more sophisticated beam-delay sparse estimation algorithms, the proposed method is much simpler to implement while still exploiting multiple structural properties of the channel.

IV. Proposed Adaptive Delay Window + Soft Low-Rank Denoising Algorithm

The main objective of the proposed algorithm is to improve the quality of the noisy channel estimate by exploiting two

fundamental structural properties of wireless MIMO channels: limited delay spread and approximate low-rank spatial structure. The proposed method extends the baseline delay-domain filtering approach by additionally performing spatial low-rank denoising within the adaptively selected delay support.

The algorithm operates in two domains: the delay domain, where temporal sparsity is exploited, and the spatial domain, where the channel matrix structure is analyzed using singular value decomposition (SVD). By combining these two denoising mechanisms, the algorithm removes noise that cannot be eliminated by delay filtering alone.

1. Motivation for the algorithm:

Before describing the algorithm in detail, it is useful to understand why the proposed approach is expected to improve channel estimates.

In the considered MIMO-OFDM system, the channel tensor can be represented as

$$H \in \mathbb{C}^{T \times K \times M \times N}$$

where:

- T represents time samples,
- K represents OFDM subcarriers,
- M represents user equipment antennas,
- N represents base station antennas.

For each time-frequency sample (t, k) the channel matrix

$$H_{\{t, k\}} \in \mathbb{C}^{M \times N}$$

describes the spatial propagation channel between the base station and the user.

When the channel estimate is corrupted by noise, two types of distortions appear:

1. Frequency-domain noise, which spreads across all subcarriers.
2. Spatial noise, which contaminates the weaker singular modes of the channel matrix.

To address this limitation, the proposed method first estimates the dominant delay support adaptively from the noisy power delay profile after noise-floor correction. It then applies soft low-rank shrinkage to the spatial channel matrices within the retained delay support. This approach provides an effective compromise between performance and computational complexity.

Compared to more sophisticated beam-delay sparse estimation algorithms, the proposed method is much simpler to implement while still exploiting multiple structural properties of the channel.

2. Step 1: Transformation to the Delay Domain

The first step of the algorithm converts the channel representation from the frequency domain to the delay domain.

This transformation is performed by applying an inverse Fourier transform across the subcarrier dimension:

$$Z = \text{IFFT}(H_{\text{noisy}})$$

In practice, this operation is applied along the subcarrier axis of the channel tensor.

The delay-domain representation Z corresponds to the **channel impulse response**. In this domain, multipath components appear as peaks at different delay taps corresponding to the physical propagation delays of reflected signals.

Due to the finite propagation delay of multipath components, the channel impulse response typically contains significant energy only within a small number of delay taps.

Therefore, the delay-domain representation is **sparse**, meaning that most delay coefficients contain only noise.

3. Step 2: Adaptive Delay Support Estimation

After transforming the channel to the delay domain, the next step is to estimate the dominant delay support adaptively from the observed power delay profile (PDP).

The average PDP is first computed across time and antenna dimensions. Since noise raises the PDP floor across many delay bins, a robust noise-floor estimate is obtained using the median PDP value. This estimated noise floor is then subtracted from the PDP, and negative values are clipped to zero.

The denoising window is selected as the smallest circular delay window that captures a prescribed fraction of the cleaned PDP energy. In this way, the selected support is data-driven and adapts to the observed channel realization.

To avoid including excessive noise or neighboring weak delay components, the maximum allowed window length is constrained. This prevents the adaptive window from becoming excessively wide in moderate-SNR conditions.

Step 3: Spatial Soft Low-Rank Denoising

For each retained delay tap, the channel contains a spatial matrix describing the propagation between antennas. Even within the valid delay support, these spatial channel matrices may still contain residual noise.

To suppress these components, singular value decomposition (SVD) is applied to each retained spatial channel matrix:

$$H = U\Sigma V^H$$

where U and V contain the left and right singular vectors, and Σ contains the singular values.

Instead of applying hard rank truncation, the proposed method uses **soft singular-value shrinkage**. Small singular values are more likely to be noise-dominated and are therefore attenuated more strongly, while dominant singular values are preserved.

This approach is more flexible than hard rank selection because it reduces noise without abruptly removing weak but potentially useful spatial modes.

Step 4: Selection of Adaptive Denoising Parameters

The final algorithm uses an adaptive delay-window selection rule together with soft singular-value shrinkage. The window is estimated from the noise-floor-corrected power delay profile, while additional SNR-aware constraints are used to prevent the selected support from becoming excessively wide.

At lower uplink SNR, larger windows and stronger shrinkage are used to improve robustness against noise. At higher uplink SNR, tighter windows and weaker shrinkage are used to preserve more of the channel structure. These choices were motivated by the observed trade-off between noise suppression and signal distortion in the spectral-efficiency experiments.

7. Summary of the Algorithm

The proposed channel denoising method can be summarized as follows:

1. Transform the noisy channel to the delay domain using an inverse FFT.
2. Compute the average power delay profile and estimate its noise floor.
3. Select an adaptive circular delay window from the cleaned PDP.
4. For each retained delay tap, perform singular value decomposition.

5. Apply soft shrinkage to the singular values in order to suppress weak noise-dominated modes.
6. Reconstruct the denoised spatial channel matrices.
7. Transform the denoised channel back to the frequency domain.

V. Experimental Results and Performance Analysis

In this section, the performance of the proposed channel denoising algorithm is evaluated and compared against the baseline hard-window method. The evaluation focuses on the achievable spectral efficiency when the precoder is designed using different channel estimates.

The following four channel representations are considered:

1. **Ideal channel** precoder computed using the true channel matrix.
2. **Noisy channel** precoder computed directly from the noisy estimate.
3. **Hard-window baseline** channel denoised using delay-domain filtering.
4. **Proposed method** delay-domain filtering followed by spatial low-rank denoising.

The downlink SNR is fixed at **5 dB**, while the uplink estimation SNR varies among:

$$SNR_{UL} \in \{-20, -15, -10\} \text{ dB}$$

For each scenario, the instantaneous spectral efficiency is computed for all time–frequency samples and the empirical cumulative distribution function (CDF) is obtained.

VI. 1. Power Delay Profile

The average power delay profile (PDP) is shown in Fig. X.

The PDP illustrates how the channel energy is distributed across delay taps after transforming the channel to the delay domain. The results show that the majority of the channel energy is concentrated within a small number of delay bins near the beginning of the delay axis. Most of the remaining delay bins contain negligible power.

This observation confirms that the wireless channel exhibits **limited delay spread**, which justifies the use of delay-domain filtering techniques such as the hard-window method. By removing taps outside the dominant delay region, a significant portion of the estimation noise can be suppressed.

However, the PDP also indicates that noise may still remain within the valid delay taps. Therefore, additional denoising mechanisms are required to further improve the channel estimate.

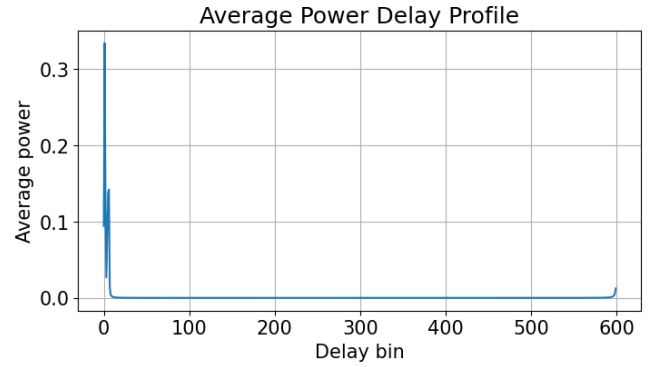


Fig 1

VI. 2. Beam-Domain Power Spectrum

The beam-domain power spectrum is shown in Fig. 2.

To obtain this representation, the spatial dimension of the channel is transformed using a spatial Fourier transform. The resulting spectrum illustrates how the channel energy is distributed across spatial beams.

The results reveal that the channel energy is highly concentrated in a limited number of beam indices. In particular, a few beams exhibit strong power peaks, while most beams contain very little energy.

This behavior indicates that the propagation environment is dominated by a small number of spatial directions corresponding to dominant multipath components. Such directional sparsity is a well-known property of wireless channels, particularly in systems with large antenna arrays.

The beam-domain concentration suggests that the spatial channel matrix should exhibit **strong correlations between antennas**, which naturally leads to approximate low-rank behavior.

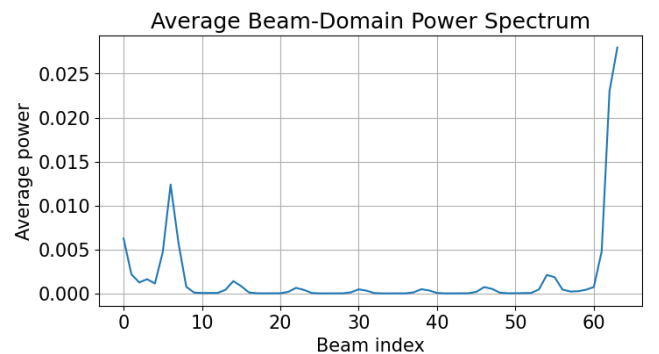


Fig 2

VI. 3. Singular Value Distribution

The average singular values of the channel matrices are shown in Fig. 3.

The singular value decomposition of the channel matrix reveals the strength of different spatial propagation modes. The results show a rapid decay in singular values, with the first

two singular values capturing the majority of the channel energy.

Specifically:

- The first singular value represents the strongest spatial propagation mode.
- The second singular value remains significant but is smaller than the first.
- The remaining singular values decrease rapidly.

This behavior indicates that the channel matrices can be approximated effectively using a small number of dominant spatial modes. Consequently, weak singular components are likely dominated by noise and can be removed through low-rank approximation.

This behavior indicates that the channel matrices can be approximated effectively using a small number of dominant spatial modes. Consequently, weak singular components are likely dominated by noise and can be suppressed through soft singular-value shrinkage. This observation supports the use of low-rank denoising in the proposed algorithm.

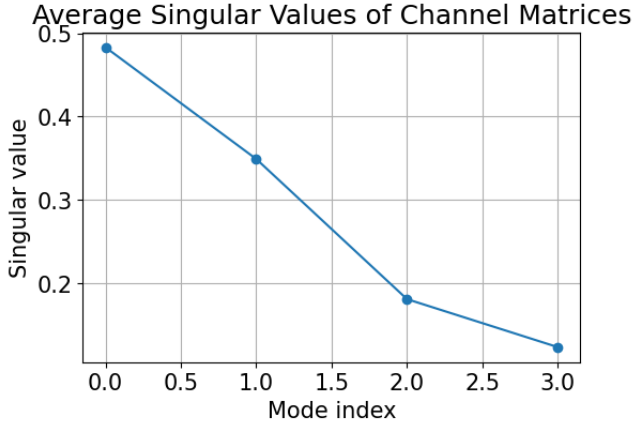


Fig 3

VI. B. Spectral Efficiency Performance

The performance of the proposed algorithm is evaluated using the instantaneous spectral efficiency metric. The empirical CDF plots illustrate the distribution of spectral efficiency values across all time–frequency samples.

The dashed horizontal line in the plots indicates the median spectral efficiency level, which is used as a summary metric for algorithm comparison.

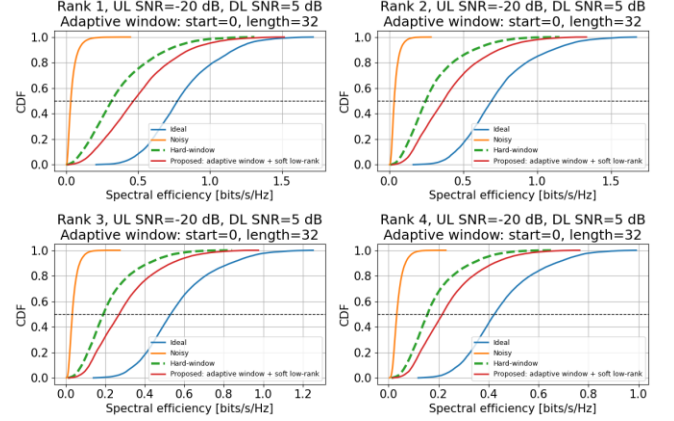


Fig 4

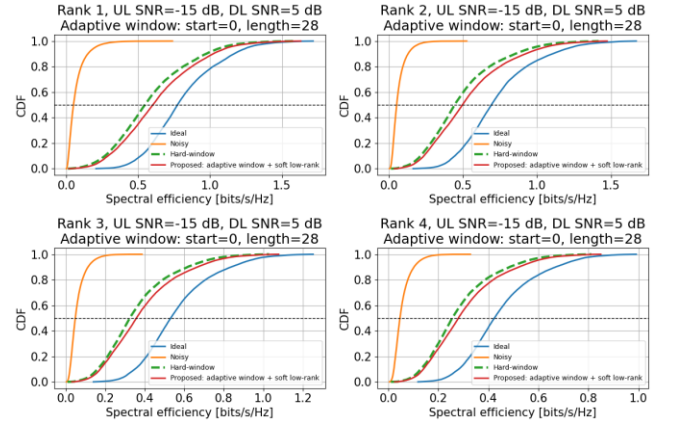


Fig 5

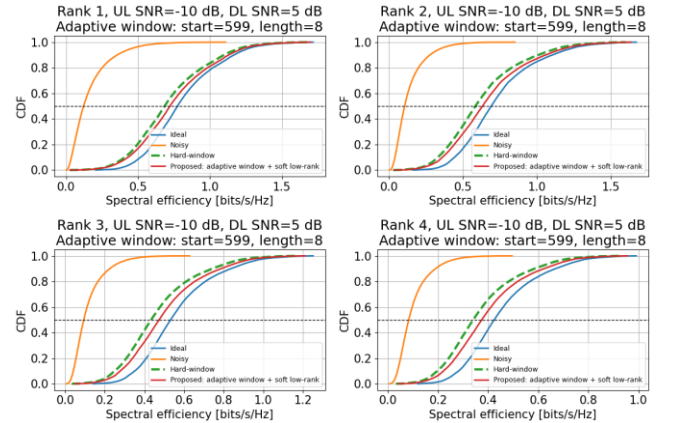


Fig 6

I. 1. Results for UL SNR = -20 dB

The scenario with **UL SNR = -20 dB** represents the most challenging channel estimation condition, where the noise level in the channel estimate is extremely high.

Under this condition, the precoder derived from the noisy channel estimate performs very poorly. The spectral efficiency values remain close to zero because the beamforming directions derived from the noisy channel are highly inaccurate.

Applying the **hard-window delay filtering** significantly improves the performance. By removing delay-domain coefficients outside the physical delay spread of the channel, the baseline method suppresses a large portion of the noise while preserving the dominant multipath components.

The proposed adaptive delay-window plus soft low-rank denoising algorithm provides additional improvements over the hard-window baseline. The soft singular-value shrinkage suppresses weak singular components that are primarily dominated by noise, resulting in a more accurate channel representation.

The median spectral efficiency values for this scenario are shown below.

Rank	HardWindow	Proposed	Improvement
1	0.314371	0.466915	+0.152544
2	0.246953	0.363909	+0.116956
3	0.190822	0.269763	+0.078941
4	0.156596	0.214701	+0.058105

The results show that the proposed method consistently improves spectral efficiency across all transmission ranks. The largest improvement occurs for rank-1 transmission, while the gain gradually decreases for higher ranks because higher-rank transmission relies on weaker spatial modes that are more sensitive to noise.

These results confirm that the additional low-rank denoising stage effectively removes spatial noise components that remain after delay filtering.

II. 2. Results for UL SNR = -15 dB

When the uplink SNR increases to **-15 dB**, the quality of the channel estimates improves compared to the previous scenario. As a result, the noisy-channel precoder achieves slightly higher spectral efficiency, although its performance is still far below that of the ideal channel.

The hard-window delay filtering again produces a substantial performance improvement by exploiting the limited delay spread of the wireless channel.

The proposed algorithm continues to provide additional improvements over the baseline method. However, the performance gains become smaller compared to the -20 dB scenario because the channel estimate already contains less noise.

The median spectral efficiency values are summarized below.

Rank	HardWindow	Proposed	Improvement
1	0.545588	0.600248	+0.054660
2	0.442340	0.491829	+0.049489
3	0.325021	0.356638	+0.031617
4	0.255865	0.279297	+0.023432

The results indicate that the proposed method remains beneficial in moderate noise conditions. The improvements are smaller than in the -20 dB case but remain consistent across all transmission ranks.

III. 3. Results for UL SNR = -10 dB

At UL SNR = **-10 dB**, the channel estimate becomes significantly cleaner than in the lower-SNR scenarios, and the hard-window baseline already performs well. Nevertheless, the proposed adaptive delay-window plus soft low-rank denoising method still provides clear additional gains across all transmission ranks.

Unlike the earlier hard-rank truncation version, the final method avoids over-suppressing weak but useful spatial modes. This is achieved by combining a tighter adaptive delay support with soft singular-value shrinkage rather than abrupt rank truncation. As a result, the proposed method maintains a consistent advantage over the hard-window baseline even at the highest tested uplink SNR.

Rank	HardWindow	Proposed	Improvement
1	0.683980	0.720572	+0.036592
2	0.584551	0.632455	+0.047904
3	0.433226	0.473537	+0.040311
4	0.339656	0.374198	+0.034542

VI. Why These Parameters were chosen

Practical Choice of Adaptive Parameters:

In the final implementation, the denoising parameters were selected in an SNR-aware manner. At very low uplink SNR, wider delay windows and stronger singular-value shrinkage were used to increase robustness against severe noise. At higher uplink SNR, tighter delay windows and weaker shrinkage were used to preserve more channel detail and avoid unnecessary suppression of useful signal components.

These choices are physically motivated by the trade-off between noise suppression and signal preservation. At low

SNR, noise dominates the LS estimate and stronger denoising is beneficial. At higher SNR, the channel structure is already more visible, so overly aggressive denoising can remove useful information. Therefore, the final algorithm uses stronger denoising in the low-SNR regime and milder denoising in the moderate-SNR regime.

Strengths and Weaknesses of the Proposed Method

Strengths

The proposed method

- Exploits both delay-domain sparsity and spatial low-rank structure, which leads to stronger denoising than delay filtering alone.
- The adaptive window makes the algorithm more robust to channel variations than a fixed delay support.
- Soft singular-value shrinkage is more flexible than hard rank truncation and reduces the risk of removing weak but useful channel modes.
- The method remains computationally much simpler than more sophisticated beam-delay sparse recovery methods while still delivering consistent gains over the baseline.

Weaknesses

- The method still relies on heuristic parameter tuning, particularly through SNR-aware settings for window size and shrinkage strength.
- The use of singular value decomposition for every retained delay bin increases computational complexity compared to the simple hard-window baseline.
- Accurate delay-support estimation may still be difficult in extremely noisy conditions.
- The approximate low-rank assumption may be less accurate in rich-scattering environments with many equally important spatial modes.

IV. 4. Overall Performance Trends

Several important observations can be drawn from the experimental results.

First, precoders designed directly from noisy channel estimates perform extremely poorly, particularly at low uplink SNR values. This highlights the strong sensitivity of precoding algorithms to channel estimation errors.

Second, delay-domain hard-window filtering provides a substantial performance improvement by exploiting the limited delay spread of wireless channels and removing noise outside the dominant delay support.

Third, the proposed **adaptive delay-window plus soft low-rank denoising** method consistently outperforms the hard-window baseline across all tested uplink SNR conditions. By combining adaptive delay-support selection with soft singular-value shrinkage, the method suppresses both delay-domain noise and residual spatial noise within the retained delay taps.

Fourth, the largest performance gains are observed at low uplink SNR, where channel estimates are highly corrupted by noise and additional denoising is most beneficial. As the uplink SNR increases, the channel estimates become more accurate and the relative gain over the baseline becomes smaller, although the proposed method still maintains a consistent advantage.

Finally, higher transmission ranks remain more sensitive to channel estimation errors because they rely on weaker spatial modes of the channel matrix. However, the use of **soft** low-rank shrinkage instead of hard rank truncation reduces the risk of removing weak but useful channel components, which helps preserve performance even at higher SNR.

Overall, the results demonstrate that jointly exploiting delay-domain sparsity and approximate spatial low-rank structure provides an effective denoising strategy for MIMO channel estimation. The proposed method is especially beneficial in challenging low-SNR scenarios, while still maintaining improvements in moderate and higher SNR conditions.

V. Conclusion

This work investigated channel denoising for MIMO-OFDM precoding under noisy uplink channel estimation. A baseline hard-window method was first evaluated, and then an improved denoising method based on adaptive delay-window selection and soft low-rank shrinkage was proposed.

The results showed that the proposed method consistently improves the spectral efficiency obtained with the hard-window baseline across all tested uplink SNR values. Structural analysis using the power delay profile, beam-domain spectrum, and singular value distribution confirmed that the channel exhibits limited delay spread, directional concentration, and approximate low-rank behavior.

Overall, the study demonstrates that jointly exploiting delay sparsity and spatial low-rank structure provides an effective and computationally practical denoising strategy for MIMO channel estimation.